

PETROLEUM PRODUCTION ENGINEERING

MODULE-16

OFFSHORE PRODUCTION SYSTEM FIXED AND FLOATING PRODUCTION

By

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Introduction

- Offshore Engineering is the engineering discipline that deals with the design and construction of structures intended to work in a stationary position in the ocean environment



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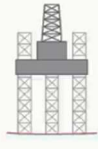
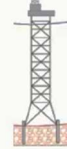
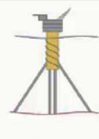


Offshore Production System

- An offshore platform is a large structure designed to stand on the seabed to provide a stable base above water for drilling and production of oil and gas wells.
- It has facilities to drill wells, to extract and process oil and natural gas, and/ or to temporarily store product before it can be brought to shore for refining and marketing.
- In many cases, the platform contains facilities to house the workforce also.

Types of Offshore Production System

- Offshore platforms are classified either as **Bottom-Supported** or **Floating**.

BOTTOM-SUPPORTED	Fixed			Compliant	
					
	Jacket	Gravity-Based Structure	Jack-up	Compliant tower	
FLOATING	Neutrally buoyant				Positively buoyant
					
	Ship FPSO	Semi-Sub	Sevan FPSO	Spar	Tension Leg Platform (TLP)

Bottom-Supported: Offshore Production System

- **Fixed Jacket Type**
 - Anchored directly into the seabed, fixed platform rigs consist of a tall, steel structure known as a "jacket".
 - Economical for water depth up to 1500 ft.



Concrete Steel/
Gravity Platform



Jackup Platform



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Bottom-Supported: Offshore Production System

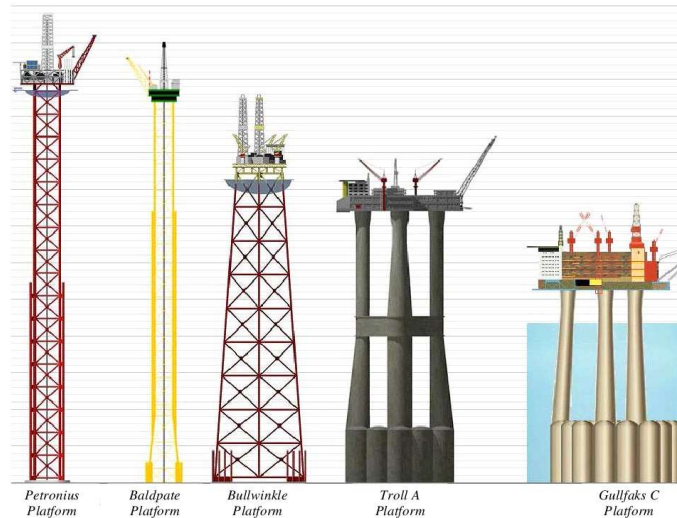
- **Compliant Structure Type Platform**
 - It is anchored to the seabed and holds most of their equipment above the surface.
 - But unlike fixed platforms, compliant towers are taller and narrower, and they sway with the wind and water almost as if they were floating.
 - Their jackets are broken into two or more sections, with the lower part serving as the base for the upper jacket and surface facilities.
 - Operates at greater depths than fixed platforms, potentially up to 3,000 feet below the surface.

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Bottom-Supported: Offshore Production System



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Floating: Offshore Production System

- These have a floating structure moored to the seabed.
- Depending upon structure and mooring, there are various types.
- Some are mentioned below:
 - Tension leg platform
 - Semi-submersibles
 - FPSO
 - SPAR platform
 - Tanker based system FPS.

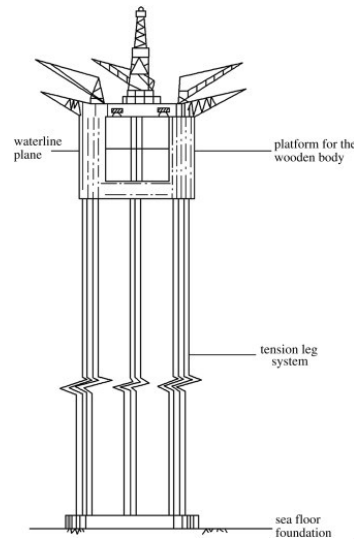
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Tension leg platform

- In this type, a large hollow section hull structure similar to a semi-submersible is held in place with many high strength steel teethers at each corner.
- The hull structure provides high buoyancy and keeps the teethers under tension.
- This is a vertically moored platform without heave, pitch and roll motions and surge, sway and tilt.



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Semi-Submersibles

- A typical Semi-submersible design has four columns connected at the bottom by pontoon with a nominally rectangular cross-section.
- A truss structure connects the column tops and supports topsides modules.
- This arrangement provides a high degree of flexibility in fabrication methodology.
- These columns are supported by two parallel pontoons or a ring pontoon.
- Pontoons are sized to provide adequate buoyancy to support all weights and vertical loads, and proportioned to maximize heave damping.

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FPSO/FSO

- Floating production, storage, and offloading systems; offloading of the crude oil to a shuttle tanker; these are typically converted or newly built tankers that produce and store hydrocarbons, which are subsequently transported by other vessels to terminals or deepwater ports.
- Floating storage and offloading system; like the FPSO, these are typically converted or newly built tankers.
- They differ from the FPSO by not incorporating the processing equipment for production; the liquids are stored for shipment to another location for processing.

SPAR platform

- SPAR Platform (SPAR) consists of a large diameter single vertical cylinder supporting a deck.
- It has a typical fixed platform topside (surface deck with drilling and production equipment), three types of risers (production, drilling, and export), and a hull which is moored using a taut catenary system of six to twenty lines anchored into the seafloor.
- SPAR's are presently used in water depths up to 8000 feet, although existing technology can extend its use to water depths as great as 9500 feet.

Tanker based system FPS

- Floating production systems; universal term to refer to all production facilities that float rather than are structurally supported by the seafloor; included would be TLP's, spars, semisubmersibles, shipshape vessels, etc.
- The term is also frequently used to describe the general category of floating production facilities that do not have onsite storage.
- The term is also used by the American Bureau of Shipping to describe a classification of floating production facilities that do not have storage capability.